

REAL TIME EVENT SYNCHRONIZATION ENGINE

A React JS based real-time event ticket booking system with synchronization concept (simulated using state updates). Includes event listing, booking, and live updates.

```
// App.js
import React, { useState, useEffect } from "react";
import "./App.css";

const eventsData = [
  { id: 1, name: "Tech Fest", seats: 50 },
  { id: 2, name: "AI Seminar", seats: 30 },
  { id: 3, name: "Workshop", seats: 20 }
];

function App() {
  const [events, setEvents] = useState(eventsData);
  const [selectedEvent, setSelectedEvent] = useState(null);

  const bookSeat = (id) => {
    const updated = events.map(event =>
      event.id === id && event.seats > 0
      ? { ...event, seats: event.seats - 1 }
      : event
    );
    setEvents(updated);
  };

  // Simulate real-time sync
  useEffect(() => {
    const interval = setInterval(() => {
      setEvents(prev =>
        prev.map(e => ({
          ...e,
          seats: Math.max(e.seats - Math.floor(Math.random() * 2), 0)
        }))
      );
    }, 5000);

    return () => clearInterval(interval);
  }, []);

  return (
    <div className="App">
      <h1>Real Time Event Synchronization Engine</h1>

      <div className="events">
        {events.map(event => (
          <div key={event.id} className="card">
            <h3>{event.name}</h3>
            <p>Available Seats: {event.seats}</p>
            <button onClick={() => bookSeat(event.id)}>
              Book Seat
            </button>
          </div>
        ))}
      </div>
    </div>
  );
}

export default App;

// App.css
.App {
  text-align: center;
}
```

```
padding: 20px;
}

.events {
  display: flex;
  justify-content: center;
  gap: 20px;
}

.card {
  border: 1px solid #ccc;
  padding: 20px;
  width: 200px;
}
```